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specifications devised and sent to hardware and software developers individually, cause extensive redesign any time when a change is required. It lacks a unified hardware-software representation, which leads to difficulties in verifying the entire system and hence incompatibilities across hardware/software boundaries. A prior definition of partitions also leads to sub-optimal designs."

Need for co-design and co-verification

Co-design and co-verification refer to parallel or concurrent development of hardware and software for a systems-on-chip (SoC) or an embedded product. The definition varies from company to company depending on the scope of the information presentation. However, the basics remain the same, i.e., simultaneous design and verification of hardware and software blocks in a system.

"The concept of hardware and software co-design can be extended from an SoC design to entire embedded system design," says Arya.

"In an embedded SoC system, co-design stands for concurrent design encompassing hardware portion of the design, typically implemented via register-transfer-level (RTL) code and embedded software contents. Today, there is almost no chip with only hardware content. Virtually all chips require some level of software content (drivers, operating systems and applications) to function according to specifications," shares Moorthy.

Justifying this with an instance, Muthu Raman Iyer, key accounts manager, FTD Infocom, shares, "The common practice today is to incorporate a very simple, non-parametric model of micro-electromechanical system (MEMS) device to verify its mixed-signal behaviour with the ICs. It is tedious for MEMS engineers to manually create models that can be used by their counterparts in IC design. When a MEMS design changes, they have to repeat the model creation effort."

Mallikarjuna B.S, senior application engineering manager from Mentor Graphics, shares, "Co-design and co-verification reduce the overall design and development cycle of your embedded system. It helps in finding the bugs at an early stage and reduces the number of errors/bugs at the system level, particularly those at the hardware-software interface level. It also helps the designers to do software validation before the hardware prototype is ready, which means accelerating the design closure time. Often, co-verification also helps in making appropriate decisions about hardware-software partitioning to optimise the throughput and efficiency of the system."

"Semiconductor companies are under tremendous pressure to deliver a fully functional chip to the market within shorter time windows to ensure profitability. If the company misses the profitability window due to a defective chip, or one that has to go through several re-spins, it translates into a huge loss in investment for the company," says Arya. "The advantages achieved through co-verification largely centre around maximising the productivity of designers and ensuring predictability in the designs, thus ensuring quicker time-to-market."

Co-design flow for embedded system

In an embedded system design, after the requirements are defined, these are partitioned into hardware and software activities. "When these activities are taken up as independent processes, they yield locally optimised designs which give rise to surprises and mismatch at the integration phase," cites Gupta. "Also, since the activities are distributed to separate and non-overlapping pure hardware and pure software teams, debugging during the integration stage becomes a nightmare with people being clueless about where the system is breaking down," he adds.

Hardware-software co-design helps